

Deepak Mallampati

FORWARD DEPLOYED ENGINEER / APPLIED AI

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Remote US / Kent, OH -- F-1 OPT, authorized to work now

PROFESSIONAL SUMMARY

AI engineer (4+ years) with production agentic + RAG systems and Kubernetes/cloud-native MLOps experience. Shipped **25+ AI agents** and **12 APIs/microservices** at Vanguard serving **100K+ requests/day** across **20+ internal teams** at **800ms p95**; built fault-tolerant automation infrastructure with exponential-backoff retry at Progress Solutions that cut failed production runs by **60%**. Customer-embedded delivery posture: multi-stakeholder integration, observability-first reliability, and willingness to ramp on HPC-specific stacks (SLURM, CRIU, kernel-layer scheduling) for the right enterprise deployment.

CORE COMPETENCIES

Customer-Embedded Engineering Delivery . Production AI Agents & LLM Orchestration . Kubernetes / Docker / Cloud-Native MLOps . Reliability & Observability Engineering . Python / FastAPI / Microservices . GPU Workload Familiarity (QLoRA on H100) . Multi-Stakeholder Integration . End-to-End Production Ownership

EXPERIENCE

Progress Solutions

Jul 2025 -- Present

AI Engineer -- USA | Healthcare AI

- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%** -- the cheapest reliability lever for customer-facing systems.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, reducing token consumption **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination.
- Established a **LLM evaluation and monitoring framework** (RAGAS, BERTScore, latency/accuracy dashboards) -- surfaced regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing while sustaining SLA targets.

Vanguard

Jan 2024 -- Jan 2025

AI Engineer Intern -- USA | Regulated FinTech

- Developed and shipped **25+ AI agents** and workflows for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across multiple business units.
- Built **12 APIs and microservices** powering new AI capabilities -- reduced p95 latency from 1.5s to **800ms** and accelerated feature integration by **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across **20+ internal teams**.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini** across the same task suite, improving task success rates by **27%** on internal evaluation datasets.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%**.

Kent State University

Oct 2023 -- Jan 2024

ML & Gen AI Research Assistant -- Ohio, USA | Research Computing

- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated instruction dataset -- direct GPU-workload experience including memory/bandwidth tradeoffs.
- Built a **privacy-preserving ML pipeline** on a clinical readmission dataset, applying k-anonymity, l-diversity, and Laplace differential privacy; reduced re-ID risk from 3.38% to 0.32% while retaining 99% baseline predictive performance.
- Authored a peer-review-ready **IEEE-format conference paper** documenting methodology and results.

Emerson

Jun 2022 -- Apr 2023

ML Trainee -- India | Industrial ML

- Developed and trained supervised regression/classification models on operational pipeline-storage data for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction on domain-specific financial datasets, achieving **89% F1-score** on custom NER tasks.

PROJECTS

career-ops

Autonomous AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision. Conductor-subagent orchestration with observability and retry -- direct analog to customer-embedded deployment automation.

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) integrating multiple AI vision providers across **29+ sites**. Demonstrates fast solo ship from problem to production.

EDUCATION

M.S. in Computer Science -- Kent State University, Ohio

2025 | GPA 3.70

TECHNICAL SKILLS

Languages & Frameworks: Python (primary), TypeScript, Java, SQL | FastAPI, React | PyTorch, TensorFlow

Cloud-Native & MLOps: Docker, Kubernetes | AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure, GCP Vertex AI | Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, W&B | observability/logging

GPU & Inference: NVIDIA H100 (QLoRA fine-tune), GPU-backed LLM inference, semantic caching, model routing, prompt compression for SLA targets

Production AI Systems: Agentic LLM orchestration, RAG (FAISS, Pinecone, Weaviate), structured outputs, evaluation pipelines (RAGAS, BERTScore), guardrails

Reliability & Integration: REST/microservices, fault tolerance + exponential backoff, third-party API integration, multi-stakeholder partnership across data + product + platform teams

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required